BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: NOTE: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Marks

Q1. 1. Choose the correct Answer for each from the given question:

Q1. According to Avogadro, one-gram mole of any gas at S.T.P. occupies volume:

(A) 22.4 mm3
(B) 22.4 cm3
(C) 22.4 dm3
(D) 22.4 m3

Q2. Dry cell produces a potential of:

(A) 1.0 volt
(B) 1.5 volt
(C) 2.0 volt
(D) 12 volt

Q3. Leakage of Helium gas through gas balloons is an example of:

(A) Diffusion
(B) Effusion
(C) Evaporation
(D) Mobility

Q4. It is a type of dipole-dipole interaction:

(A) Covalent bond
(B) Hydrogen bond
(C) Metallic bond
(D) ionic bond

Q5. The only Halogen exists in liquid state at room temperature:

(A) F2
(B) Cl2
(C) Br2
(D) I2

Q6. 5% m/m aqueous solution means that 5 g of solute is dissolved in:

(A) 90 g of water
(B) 95 g of water
(C) 100g of water
(D) 105 g of water

Q7. The branch of Chemistry that deals with the compounds in living organisms is called as:

(A) Analytical Chemistry
(B) Bio Chemistry
(C) Organic Chemistry
(D) Physical Chemistry

Q8. Electrons pairs which are not shared by atoms, are called:

(A) electron pairs
(B) lone pairs
(C) bond pairs
(D) shared pairs

Q9. The symbol of ‘silver’ is:

(A) Ag
(B) Au
(C) Sl
(D) Sn

Q10. This atomic model is known as the “quantum mechanical model” of the atom:

(A) Bohr's atomic model
(B) Rutherford atomic model
(C) Dalton's atomic model
(D) Schrodinger atomic model

Q11. Fourth and fifth periods in periodic table, are called:

(A) short period
(B) long period
(C) normal period
(D) very long period

Q12. Electronic configuration is the distribution of:

(A) Proton
(B) Positron
(C) Neutron
(D) Electron

SECTION “B” (Short Answer Question)

Maximum Marks: 24

Note: NOTE: Answer any EIGHT questions from this section. All questions carries Three (3) marks.

Q2. What is meant by the terms 'period' and 'group' with respect to periodic table. Identify period and group of 7N and 11Na

Q3. Calculate the number of moles and number of molecules present in 45 grams of glucose (C6H12O6).

Q4. Write down the electronic configuration of the following elements:

(i) Carbon

(ii) Fluorine

(iii) Sodium

Q5. Define oxidizing and reducing agents. Identify oxidizing and reducing agents of the following chemical reaction. $Zn + Cl_2 \rightarrow ZnCl_2$

Q6. Balance the following chemical equations by inspection method:

(i) $KNO_3 \rightarrow KNO_2 + O_2$

(iii) $CO + O_2 \rightarrow CO_2$

Q7. The pressure of a sample gas is 3 atm and the volume is 5 litre. If the pressure is reduced to 2 atm then what would be its new volume?

Q8. Write down three differences between amorphous solids and crystalline solids.

Q9. Define the following:

(i) Saturated solution

(ii) Unsaturated solution

(iii) Super saturated solution

Q10. Write down three important characteristics of covalent compounds.

Q11. How much sodium hydroxide (NaOH) is required to prepare 400cm3 of 0.3 M solution?

Q12. Write down three properties of cathode rays.

Q13. Justify the followings:

(i) Most of Ionic compounds are found in solid states.

(ii) Alkali metals form cation easily.

SECTION “C” (Detailed –Answer Question)

Maximum Marks: 24

> Note: NOTE: Attempt any FOUR questions from this Section. Each question tries (SIX) 6 marks.

Q14. Explain the corrosion of Iron and describe any four methods to prevent metals from it

Q15. Explain the importance of glues and epoxy resins in our society.

Q16. An atom has only one electron in its ‘M’ shell then what would be its atomic number? Identify that element and write four common characteristics of the group from - which it belongs to.

Q17. What is coordinate covalent bond? Explain it by the formation of Ammonium Ion (NH4+) and hydronium ion (H3O4–)

Q18. Compare the characteristics of solution, colloid and suspension,

Q19. Define Charle's law of gases and explain mathematical representation of it.